

Quality systems engineer with five years of experience building automation, CI, observability, and AI-agent infrastructure for a multi-tenant B2B platform. I turn powerful but failure-prone automation into systems teams can trust through explicit evidence rules, deterministic guardrails, executable knowledge, and production feedback loops.

VERIFIED SCALE

770+

Playwright browser tests across three roles

240+

Executable application skills for autonomous agents

1,300+

Assertions in the daily API regression program

190

Documented product flows exercised across four roles

CORE SYSTEMS

- AI reliability**
Evidence contracts, deterministic hooks, watchdog agents, progress ledgers, drift sentinels
- Test architecture**
Playwright, TypeScript, multi-role auth, fixtures, API and data-layer verification
- CI and observability**
GitHub Actions, Sentry, session replay, failure classification, production monitoring
- Languages and data**
TypeScript, JavaScript, Python, SQL, GraphQL, REST, PostgreSQL

EDUCATION

B.S. Computer Science
Fairleigh Dickinson University, 2020
National Society of Black Engineers

EXPERIENCE

Software Development Engineer in Test

Private multi-tenant B2B SaaS platform | 2024-present

- Own quality end to end as a one-person QA function: test architecture, CI reliability, API contracts, production monitoring, and release evidence.
- Built and operate a 770+ test Playwright and TypeScript browser suite across three user roles. Redesigned the main pipeline into six parallel runners with shared authentication, cutting runtime from about 3.5 hours to 1.6 hours.
- Taught an AI agent the application through 240+ versioned executable skills that capture selectors, fixtures, edge cases, and expected outcomes. Wrapped it in deterministic evidence gates and watchdogs.
- Ran a full-product autonomous regression program across 190 documented flows and four roles. The system completed without silent skips or false bug reports.
- Built a daily API regression program with 50 Postman collections and 1,300+ assertions, roughly 99% steady-state signal, automated issue reconciliation, and team reporting.
- Produced the first API coverage baseline by inventorying 1,090 endpoints across four API technologies. Extended it into a frontend-to-backend traceability tool showing 52% of reachable backend functions covered by automation.
- Found a production feature that had failed silently for two years: 2,473 recorded errors and 43 affected users in a 90-day window. The user interface had reported success while an internal call failed.
- Analyzed 104 production session replays and converted them into eight confirmed defects with source-level root causes, including a deploy-time failure affecting users with open tabs.
- Restored trust after 19 consecutive red CI days by reproducing every failure live, proving the streak contained no new product defects, and fixing environment-load and hydration races at the system level.

Quality Assurance Engineer

Private multi-tenant B2B SaaS platform | 2021-2024

- Built the company's first formal regression program and served as the release quality gate for major routing, schema, workflow, and communications launches.
- Reduced escaped defects by 30%, increased automated coverage by 25%, and cut manual regression time by 50% through repeatable release gates and automation.
- Validated backend scalability to 10,000 concurrent users with K6 load and soak tests, and verified data integrity across large Salesforce-integrated workflows with SQL.

Selected systems

INDEPENDENT PRODUCTS + ENGINEERING METHODS

OPERATING PRINCIPLES

- A guard must be able to break.
- Coverage is counted in actions, not pages.
- Never classify a failure from logs alone.
- Silence is not health. Give quiet systems a sentinel.
- Keep AI knowledge where you can audit it.

ADDITIONAL TOOLS

- Postman / Newman
- GitHub Actions
- Docker / Jenkins
- New Relic / PagerDuty
- TestRail / Jira
- K6 / JMeter
- Firebase / Apollo
- Bash automation

SELECTED BUILDS

Hallpass	Convention companion built across Next.js, Flutter, Supabase, and a layered schedule-ingestion system. Operates on roughly 2,900 conventions and 25,000+ events with active data-quality and SEO controls.
Compounding Second Brain	Automated capture, distill, and lint system for a cited, version-controlled Obsidian knowledge graph with immutable sources, federated topic wikis, and drift repair.
E7 Advisor	Verified combinatorial equipment optimizer. The pruned search matched a brute-force oracle 12 of 12 times; stat calculations matched 111 characters with zero differences.

RELIABILITY METHOD

Evidence contracts	Every verdict has an explicit proof requirement. A pass needs a real action and an observed result; logs alone cannot classify a failure.
Machine-enforced honesty	Deterministic hooks reject shallow checks, unsupported completion claims, silent skips, and false blocked verdicts at agent turn boundaries.
Durable autonomous work	Versioned skills, progress ledgers, checkpoints, and watchdogs let long-running programs survive context limits while keeping every item accountable.
Self-monitoring pipelines	Watermarks, structural validation, rolling backups, failure logs, and sentinels surface quiet decay in AI and knowledge workflows.